

Akshay Shailendra Pakhare

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akshayspakhare.github.io

Postdoctoral Research Associate, School of Engineering, Brown University (September 2023 – Present).

EDUCATION

Doctor of Philosophy, Mechanical Engineering

Michigan State University, East Lansing, MI, USA.

January 2020 - August 2023

New Jersey Institute of Technology, Newark, NJ, USA.

September 2017- December 2019

Thesis: "Interface Fracture Characterization and an Improved Multiphysics Model for Rechargeable Batteries"

Thesis Advisor: Prof. Siva Nadimpalli

Master of Engineering, Materials Engineering

Indian Institute of Science, Bangalore, Karnataka, India.

August 2015 - June 2017

Thesis: "Modeling the Electrochemical Hysteresis in SnO_2 Electrode for Lithium ion battery"

Thesis Advisor: Prof. Vijay Sethuraman

Bachelor of Technology, Metallurgical Engineering

College of Engineering Pune, Pune, Maharashtra, India.

July 2009 - May 2013

Thesis: "Severe Plastic Deformation by Equal Channel Angular Pressing of Al-Si alloy"

Thesis Advisor: Prof. Santosh Hosmani

RESEARCH INTEREST

Experimental and theoretical investigation of liquid and solid-state battery materials, focusing on the mechanics of interfaces, fracture, and dendrite formation. Development of coupled electro-chemo-mechanical models and in situ techniques to understand degradation and failure in thin-film and composite electrode systems.

PROFESSIONAL RESEARCH EXPERIENCE

Postdoctoral Research Associate Advisor: Prof. Pradeep Guduru and Prof. Lucas Caretta at Brown university.

September 2023 -

Present.

- **Measured** fracture toughness of porous sulfide solid electrolytes and linked it to dendrite growth
- **Synthesized** hard-carbon and composite thin-film electrodes for simultaneous electrochemical-mechanical characterization of Na- and Li-ion systems.
- **Fabricated** dense epitaxial LLTO/LLZTO thin films to probe strain-controlled degradation in solid electrolytes.
- **Mentored** 3 undergraduate researchers → 2 honors thesis; co-supervised one interdisciplinary project on biomedical drug delivery needle design.

Research Assistant Advisor: Dr. Siva Nadimpalli at Department of Mechanical Engineering, NJIT and MSU. September 2019 - August 2023.

- **Quantified** binder/particle interface fracture using a custom blister-interferometry method → published in Exp. Mech. (2023), J. Appl. Mech. (2024), ACS App. Energy Materials (2025)
- **Modeled and Implemented** a coupled large-deformation electro-chemo-mechanical ABAQUS UEL capturing transient Butler-Volmer kinetics.

- **Led** a subgroup of 4 students; trained peers in mechanical testing and continuum modeling workflows.

TEACHING EXPERIENCE

Teaching Assistant ME 215: Engineering Materials and Processes — New Jersey Institute of Technology
September 2017 - June 2019.

- Led weekly lab sections (≈ 20 students) and designed hands-on experiments on material testing and processing.
- Graded lab reports and assignments; provided feedback emphasizing experimental design and data interpretation.

Teaching Assistant ME 481: Mechanical Engineering Design Projects — Michigan State University
January 2020 - April 2020.

- Organized and evaluated senior capstone design presentations for 5 student teams.
- Guided students through project scoping, technical report preparation, and prototype development.

Honors Thesis Reader Brown University School of Engineering 2024 – Present

- Served as reader for undergraduate honors theses; provided feedback on research design and data analysis.

Undergraduate Research Mentor Michigan State University and Brown University 2020 – Present

- Mentored 7 undergraduate researchers on battery mechanics projects; two received departmental honors recognition.

MENTORSHIP

- Sam Andrade, High School Student, Summer 2026
- Lauren Lane, B.S. Materials Engineering Major, Spring 2026 - Present
- David Adesina, B.S. Materials Engineering Major, Fall 2024 - Spring 2026
- Anna Manager, B.S. Electrical Engineering Major, Fall 2024 - Spring 2025
- Chris Shin, B.S. Biomedical Engineering Major, Spring 2024 - Summer 2024
- Gavin Watthayu, B.S. Mechanical Engineering Major, Fall 2022 - Spring 2023
- Cole Treece, B.S. Mechanical Engineering Major, Fall 2022 - Spring 2023
- Bryce Hitchcock, B.S. Mechanical Engineering Major, Spring 2022 - Spring 2023
- Michael Wichman, B.S. Mechanical Engineering Major, Spring 2021 - Spring 2022

PUBLICATIONS

Rakshit, Subhajit, **Akshay S. Pakhare**, Olivia Ruiz, M. Reza Khoshi, Eric Detsi, Huixin He, Vijay A. Sethuraman, and Siva PV Nadimpalli. "Measurement of volume changes and associated stresses in Ge electrodes due to Na/Na⁺ redox reactions." *Journal of the Electrochemical Society* 168, no. 1 (2021): 010504.

Akshay S. Pakhare and Siva P.V. Nadimpalli, "Characterization and Prediction of Polymer/Active Material Interface Failure in Battery Electrodes", *Exp Mech* 63, 363–376 (2023). *Coverpage of Experimental Mechanics*

Akshay S. Pakhare and Siva P.V. Nadimpalli, "Measurement of Binder/Active Material Interface Fracture in Rechargeable Batteries and the Effect of Oxide Coating", *J. Appl. Mech.* Apr 2024, 91(4): 041009

Akshay S. Pakhare, Karun Kalia, Siva PV Nadimpalli, Amir Ameli, "Interface fracture characterization of 3D-printed rigid/flexible dissimilar polymers", *Prog Addit Manuf* 9, 2235–2247 (2024)

Amit Chanda, **Akshay S. Pakhare**, Abdulrahman Alfadhli, Vijay A Sethuraman, Siva PV Nadimpalli, "Real-time measurement of sodiation induced stress in hard carbon composite electrodes", Journal of Power Sources 609 (2024) 234678

Siyuan Song, Changmin Shi, **Akshay S. Pakhare**, Brian W Sheldon, Pradeep R Guduru, "Investigation of the Mechanical Properties of Porous Argyrodite Sulfide Electrolytes for All-Solid-State Batteries", ACS Applied Energy Materials, 2025, 8, 9, 5636–5644

Akshay S. Pakhare, Abdulrahman Alfadhli, Nikola Bosnjak, Shawn A. Chester, Siva Nadimpalli, "Impact of transient charge transfer kinetics on mechanics and electrochemistry of rechargeable batteries." International Journal of Solids and Structures, 2026

Akshay S. Pakhare, Gordon Waller, and Siva Nadimpalli, "Quantifying the Evolution of Binder/Active Material Interface Fracture Properties from the As-Prepared State to Cycling Conditions.", ACS Applied Energy Materials, 2025

Abdulrahman Alfadhli, **Akshay S. Pakhare**, Vijay Sethuraman, Siva Nadimpalli, "Self-discharge induced stresses and associated mechanical damage in Na-ion battery electrodes", Experimental Mechanics, 2026

Alec McCall, **Akshay S. Pakhare**, Chris Shin, Zoey Jiang, Pradeep Guduru, Anita Shukla "Microneedle puncture criteria for ex vivo bacteria infected porcine skin tissue ", *Manuscript submitted*

Akshay S. Pakhare, Naba Karan, Pradeep Guduru, "Mechanistic understanding of sodium storage from operando stress measurements in hard-carbon thin-films", *Manuscript submitted*

Manuscripts in preparation

Martina Borges, Amit Chanda, **Akshay S. Pakhare**, Siva Nadimpalli, "In Situ measurement of Solvent Diffusion and the Associated Changes in the Mechanical Properties of Polymer Binders"

Akshay S. Pakhare, Naba Karan, Pradeep Guduru, and Lucas Caretta, "Solid-electrolyte manuscript (title withheld until publication)"

Akshay S. Pakhare, David Adesina, Lauren Lane, Naba Karan, Pradeep Guduru, and Lucas Caretta, "Solid-electrolyte manuscript (title withheld until publication)"

Akshay S. Pakhare, Lauren Lane, Chenjie Gan, Naba Karan, Feng Lin, Brian Sheldon, Pradeep Guduru, and Lucas Caretta, "Solid-electrolyte manuscript (title withheld until publication)"

Alec McCall, Keven Alkhoury, **Akshay Pakhare**, Pradeep Guduru, Vikas Srivastava, Anita Shukla, "Evolution of Porcine Skin Viscoelasticity During Biofilm Formation: Experiments and Modeling"

Siyuan Song, **Akshay Pakhare**, Cameron Meyers, Brian W. Sheldon, Lucas Caretta, Greg Hirth, and Pradeep R. Guduru, "Solid-electrolyte manuscript (title withheld until publication)"

AWARDS and HONORS

- Orrin Barrett Research Fellowship September 2021 - January 2022
Awarded for doctoral research in solid mechanics and of mechanics of battery materials.
- Government of India, Graduate Aptitude Test in Engineering (GATE) fellowship for Master of Engineering August 2015 - June 2017
National fellowship awarded to top engineering graduates for M.E. study at IISc Bangalore.
- Society for Biomaterials Student Star Award, Boston Regional Symposium (2024) (co-mentored Alec McCall a Ph.D. Student in Biomedical Engineering at Brown who the award)

INTERNSHIP

- Indira Gandhi Centre for Atomic Research (IGCAR), Kalpakkam — Training Program, June-December 2014
Completed training focusing on mechanics and materials in nuclear industry.
- Indian Institute of Technology Bombay — Research Intern June–July 2012
Developed a graphical user interface for Electron Backscatter Diffraction (EBSD) orientation imaging (OIM) data visualization and post-processing.
- Automotive Foundry Internship, Pune Summer 2011
Studied metal casting processes for automotive components.

LEADERSHIP AND OUTREACH

- **Co-head**, “On the Etch” Materials Science Program, MindSpark Tech Festival, College of Engineering Pune 2011
Organized and coordinated a materials-processing outreach event attracting 100+ participants; supervised student volunteers and experimental demonstrations.
- **Joint Secretary**, Indian Institute of Metals – Student Chapter, Pune 2012
Managed academic events and industry outreach; coordinated student engagement with local foundry and materials-processing professionals.
- **Head**, Metallography Contest, Meta Vista Symposium 2012
Designed and judged competition on microstructural analysis techniques; mentored junior students in sample preparation and imaging.
- **Participant**, National Service Scheme (NSS) 2010–2011 Volunteered in community service programs and science outreach in rural schools.
- **Techno-craft Team Member** 2012
Contributed to mechanical design of “Emulator,” a motion-transfer demonstration device.

PROFESSIONAL SERVICE and AFFILIATIONS

Peer review service

- Journal of the Electrochemical Society
- Experimental Mechanics
- Mechanics of Materials
- Journal of Energy Storage

Professional Memberships

- Member, The Electrochemical Society (ECS)
- Member, Society for Experimental Mechanics (SEM)
- Member, Society of Engineering Sciences (SES)

Professional Leadership

- Chair, MS 404 Mechanics of Energy Materials, 20th U.S. National Congress on Theoretical and Applied Mechanics, Pasadena, CA, 2026
- Chair, 43 Multiphysics Behavior, 2026 Society of Experimental Mechanics (SEM) Annual Conference, Norfolk, VA, 2026

CONFERENCE PRESENTATION and INVITED TALKS

Presented over 15 talks at national and international conferences including ECS, SEM, and SES (2019–2026).

"Effect of Stress on Li-Transport in High Energy Density Materials", 236th Electrochemical Society(ECS) Meeting Conference Atlanta, GA.

"Effect of Stress on Li Transport in High Energy Density Electrode Materials", SES 2019 - Society of Engineering Sciences(SES), 56th Annual technical meeting, St. Louis, MO.

"Mechanical Behavior of Na-ion Battery Anode Material", Electrochemical Society(ECS) PRiME 2020 Conference, Virtual.

"Volume expansion induced stresses in Ge electrode due electrochemical cycling against sodium", 239th ECS (Electrochemical Society) Meeting, May 30-June 3, 2021, Digital Meeting

"Effect of Ion Size on the Stress Evolution in Ge Electrode during Electrochemical Cycling", 2021 SEM (Society of Experimental Mechanics) Annual Conference, June 14-17, 2021, Virtual

"Modeling the Stress and Electrochemical Response of Sodium Ion Battery Anode", 2021 Annual SES (Society of Engineering Sciences) Conference, October 1-29, 2021, Virtual

"In situ measurement and modeling of stress evolution in Ge electrode due to Na/Na+ redox reactions", 240th ECS (Electrochemical Society) Meeting, October 10-14, 2021, Digital Meeting

" Measurement and Modeling of the Mechanical and Electrochemical Response of Ge Electrode During Sodiation/Desodiation Cycling", ASME - IMECE® 2021 (International Mechanical Engineering Congress & Exposition), November 1 – 5, 2021, Virtual Conference, Online

"Characterization of interface failure between polymer binder and active materials for high energy density electrode materials", Society of Experimental Mechanics (SEM) Annual Conference, June 2022, Pittsburgh, PA.

"A continuum mechanics model and in situ characterization of high performance electrodes for Li and Na batteries", Society of Experimental Mechanics (SEM) Annual Conference, June 2022, Pittsburgh, PA.

"Experimental Method to Characterize the Polymer Binder/Active Material Interface Failure Behavior for High Energy Density Electrodes", 19th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM), June 2022, Austin, TX.

"A diffusion-deformation model and in situ stress measurement for high performance Li and Na battery electrodes ", 19th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM), June 2022, Austin, TX.

"A Fracture Mechanics Approach to Polymer Binder/Active Material Interface Failure Characterization for High Performance Electrodes", Society of Engineering Sciences (SES) Annual Conference, October 2022, College Station, TX

"In Situ Experiments and a Coupled Electrochemical-Large Deformation Model for Characterizing Cyclic Behavior of Battery Electrodes", Society of Engineering Sciences (SES) Annual Conference, October 2022, College Station, TX

"A continuum mechanics model coupled with transient electrochemical kinetics for high performance rechargeable batteries", Society of Engineering Sciences (SES) Annual Conference, October 2023, Minneapolis, MN

"Charactering the effect of electrochemical cycling on interface failure between composite electrode binder and active material", Society of Engineering Sciences (SES) Annual Conference, October 2023, Minneapolis,

MN

"Operando Stress Evolution in Hard Carbon Anodes–Insight into the Mechanical Degradation Induced Failure Mode in Rechargeable Sodium Ion Batteries", 245th Electrochemical Society(ECS) Meeting Conference, June 2024, San Francisco, CA.

"Microneedle puncture criteria for ex vivo bacteria infected porcine skin tissue". McCall, A.B., Society for Biomaterials 2024 Regional Symposia", September 2024, Boston, MA, **Received Society for Biomaterials Student Star award for presentation**

"Operando Stress Evolution in Hard Carbon Anodes–Insight into the Mechanical Degradation Induced Failure", Society of Experimental Mechanics (SEM) Annual Conference, June 2025, Milwaukee, WI.

"Degradation of electrolytes as failure mechanism in all solid state lithium batteries (ASSLBs)", Society of Engineering Sciences (SES) Annual Conference, October 2025, Atlanta, GA.

"Lithium Electrode Free Degradation and Fracture of Solid Electrolytes in All Solid-State Lithium-ion Batteries (ASSLBs)", Materials Research Society (MRS) Fall Meeting, December 2025, Boston, MA.

"Lithium Electrode Free Dendrite Formation and Fracture of Solid Electrolytes in All Solid-State Lithium-ion Batteries (ASSLBs)", Society of Experimental Mechanics (SEM) Annual Conference, June 2026, Norfolk, VA.

"Lithium Metal Electrode Free Dendrite Formation in Li-ion Solid Electrolytes", 20th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM), June 2022, Pasadena, CA.

PROFESSIONAL REFERENCES

Prof. Siva Nadimpalli

Associate Professor, Mechanical Engineering, Michigan State University, East Lansing, MI

Ph.D. Advisor

| <https://www.egr.msu.edu/~sivan/>

Prof. Shawn Chester

Professor, Mechanical Engineering, New Jersey Institute of Technology, Newark, NJ

Collaborator

| <https://web.njit.edu/~sac3/Home.html>

Dr. Vijay Sethuraman

Founder, Faraday Laboratory and Principal Scientist, Birla Carbon, Marietta, GA

Collaborator and Masters thesis advisor

| <http://www.nanopolitan.com/>

Prof. Pradeep Guduru

Professor, School of Engineering, Brown University, Providence, RI

Postdoctoral Advisor

- | <https://engineering.brown.edu/people/pradeep-guduru>

Prof. Lucas Caretta

Assistant Professor, School of Engineering, Brown University, Providence, RI
Postdoctoral Advisor

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| <https://lucascaretta.com/>